Ravi Ranjan Prasad

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Computer Science student with strong foundations in data structures, machine learning, and backend development. Experienced in building computer vision systems, training machine learning models, and developing REST APIs. Skilled in Python-based ML frameworks, real-time data processing, and integrating software with hardware systems. Interested in machine learning, AI systems, and backend engineering roles.

Skills

Programming Languages	Python, Java, C
Web Technologies	HTML, CSS, JavaScript
Frameworks & Libraries	NumPy, Pandas, Matplotlib, Seaborn, OpenCV, Scikit-learn, PyTorch, YOLOv8, MediaPipe, Hugging Face Transformers, Spring Boot
Backend & APIs	REST APIs, Spring Boot REST APIs
Databases	MySQL, PostgreSQL, SQL, Spring Data JPA
Developer Tools	Git, VS Code, Jupyter Notebook, Google Colab, Ollama, Maven
Core CS Concepts	Data Structures, OOP, DBMS, Computer Networks, Java Collections Framework, Generics, Stream API
Security	Spring Security, JWT Authentication, BCrypt
Testing	JUnit, Mockito, MockMvc

Projects

DisasterMesh (GitHub) — **MVVM, Google Nearby API**

Decentralized Android Communication System

- Built a **decentralized emergency communication system** enabling messaging during internet and cellular outages.
- Implemented **peer-to-peer mesh networking** using Google Nearby Connections API with automatic device discovery and multi-hop message relay.
- Developed a **risk-based SOS alert system** with severity levels and automatic alert propagation across connected devices.
- Integrated **offline maps using OpenStreetMap (osmdroid)** with real-time location tracking and radius-based alerts.
- Architected the application using **MVVM architecture** with background services managing connectivity, messaging, and location updates.
- Validated reliability across **10 Android devices** within a **25-meter** mesh network during offline testing.

Automated AI Sentry Gun (GitHub) — **Python, OpenCV, YOLOv8, Arduino**

Computer Vision + Embedded Systems

- Developed a **real-time computer vision targeting system** capable of detecting and tracking objects using YOLOv8 and OpenCV.
- Implemented **dynamic target selection and tracking** with low-latency inference for continuous object monitoring.
- Built a **face recognition training pipeline** enabling identification and lock-on to specific individuals.
- Integrated **Arduino-controlled pan–tilt servo hardware** via PyFirmata2 for smooth real-time physical tracking.
- Designed a **laser/relay engagement mechanism** triggered on confirmed target lock with manual safety override.
- Created a **Tkinter-based control interface** to manage detection modes, training workflows, and hardware interaction.

Education

Bachelors of Technology, CSE, *Haridwar University* — June 2027

Bachelor of Science, Data Science, *IIT Madras* — June 2027

Guru Gobind Singh Public School, *X : 94%, XII : 78%* — 2020, 2022

Activities

- **Runner-Up, TechSangram University-Level Hackathon** — Presented the **Automated AI Sentry Gun** project, demonstrating real-time computer vision and hardware–software integration.
- **PR Manager, Pichavaram House, IIT Madras** — Managed communications, outreach, and coordination activities for student-led initiatives and events.
- **NPTEL Certification — Introduction to Machine Learning** — Covered supervised/unsupervised learning, model evaluation, and core ML concepts.